


**Bachelor Project**

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# Search for a Higgs-like boson through the H → ZZ → 4l process in pp collisions at sqrt(s) = 8 TeV

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**Abstract**

The SM Higgs boson is predicted to decay (amongst others) into two Z bosons which can decay to an electron- anti-electron pair or an muon- anti-muon pair.

The invariant mass distribution of 4 leptons is analyzed with data from the Compact Muon solenoid (CERN). The expected distribution as established from simulation is compared to the actual data.

The p value in the 126 GeV bin is 0.00036 and the z score is 3.38 standard deviations. These results are significant enough to speak of evidence for a new particle with a rest mass of 126 GeV, but a more rigorous analysis is needed to boost the sensitivity of the analysis to claim a discovery.